Aniruddh Ballal

Software engineer building full-stack and mobile applications, with a background in scientific computing.

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EXPERIENCE

Indiassetz | Assistant Software Engineer — Mar–Oct 2026

- Built an investor/back-office platform with Next.js, FastAPI and PostgreSQL: role-based document approvals, capital-call settlement tracking and portfolio reporting.
- Migrated inherited React Native apps across framework and native-build changes; repaired authentication, navigation and third-party integrations.
- Added CRM payment-review and exception-approval workflows, revenue-attribution dashboards and spreadsheet exports.

Qwixpert Consulting | Data Analyst Intern — Jan–Mar 2026

- Analysed client datasets, automated reporting across linked Excel workbooks and prepared consulting presentations.
- Independently built supply-chain network design, demand-forecast comparison and dispatch-planning tools with maps, dashboards and spreadsheet reports.

Indiassetz | Frontend Web Development Intern — Jun–Jul 2025

- Built React/Material UI interfaces and reusable components, refining interactions and layouts with design and engineering teams.

IISc, Department of Physics | Solar Physics Intern — Jun–Aug 2024

- Modelled solar magnetic-field evolution using Python, spherical harmonics, NumPy and SciPy under Dr. Prantika Bhowmik; visualised solar-cycle behaviour.

SELECTED PROJECTS

Sunalyse Source — 2025–26
Reconstructed coronal fields across 181 Carrington rotations with a Python PFSS pipeline and Three.js viewer. Vectorisation and four-core tracing achieved a documented **7.6x speedup** over the original implementation.

Queen's Gambit 2.0 Source — 2025–26
Built file steganography using legal chess moves, Rust/WASM and Web Workers, with optional AES-256-GCM encryption. Documented 32.1 KB encoding benchmark: **195 s to 5.2 s**.

dedgram Source — 2026
Built an Instagram research toolkit with Android capture, Playwright scraping and a public atlas; stored profile/reel snapshots, source evidence and username history to investigate suspected automated accounts.

kNupdate Source — 2026
Built a Kotlin/Compose activity-sharing app with paired-device delivery and SQLite histories. A Cloudflare relay discards activity payloads after all recipients acknowledge or 30 seconds elapse.

SatLies Source — 2026
Implemented a Three.js orbital simulator with RK4 integration, comparing Earth's oblateness (J2) and lunar/solar gravity against a two-body baseline through live trajectories and separation plots.

Credex Source — 2026
Built a Next.js/Node.js trade-finance prototype linking procurement, payment records and loan review; a Python/FastAPI service uses PuLP to schedule payments under cash-flow and deadline constraints.

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, Kotlin, Rust, SQL.
Applications & data: React, Next.js, React Native, Node.js/Express, FastAPI, PostgreSQL, MongoDB, SQLite.
Scientific & tooling: Three.js, NumPy, SciPy, WebAssembly, Playwright, Git, Docker.

EDUCATION

Manipal Institute of Technology — Graduated Jul 2026
B.Tech, Information Technology | CGPA: 8.1/10

LEADERSHIP & ENGINEERING ACTIVITIES

ThrustMIT — Avionics member; PCB, telemetry and RF/GPS/IMU integration for student rocketry (2022–23).
Manipal Runners' Club — Co-organised the 2024 marathon (8,000+ participants) and supported six-week runner training.